



Amrita School of AI
COIMBATORE CAMPUS

RELEASED WEEK 9 · DUE WEEK 12

Assignment 3: Distribute It

Take a problem across nodes, choose a decomposition and defend it, then show where the network starts to win.

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RELEASED

Week 9

DUE

End of week 12

WEIGHT

1.5% of the course (assignments total 6%)

MARKS

30

WORK

Pairs

OFFERING

Odd Semester 2026-27

The task

Take a problem beyond one node with MPI. Either continue Assignment 2's kernel or choose from:

PROBLEM		COMMUNICATION PATTERN
A	2-D heat diffusion	Halo exchange, nearest neighbour
B	Distributed matrix-vector product	Broadcast plus reduction
C	Conjugate gradient solver	Halo exchange plus two Allreduce per iteration
D	Parallel sample sort	All-to-all redistribution
E	Distributed k-means	Allreduce of centroids each iteration

Part A: Design before code (7 marks)

A1. Apply Foster's PCAM explicitly. State the partition, the communication, the agglomeration and its granularity argument, and the mapping.

A2. Predict the communication-to-computation ratio as a function of P , from the geometry. For a grid problem, derive it for both 1-D and 2-D decomposition and say which you will use and why.

A3. Predict, using $T = \alpha + n/\beta$ and your Lab 6 measurements of α and β , the communication time per step at $P = 4, 16, 64$. State the P at which you expect communication to reach 50 percent of the step time.

Parts A2 and A3 must be written **before** you measure. The mark is for the prediction and its later reconciliation, not for being right.

Part B: Implement (12 marks)

B1. Correct MPI implementation. Verified against a serial reference at several rank counts, **including counts that do not divide the problem evenly**. Use 5, 6 and 7 ranks, not just powers of two.

B2. Non-blocking communication with genuine overlap: interior computation between the posts and the wait. Show the measurement with and without overlap.

B3. Use collectives where collectives belong. A loop of `MPI_Send` to rank 0 where an `MPI_Reduce` would do loses marks.

B4. Instrument communication and computation separately, and report the communication fraction at each rank count.

B5. Handle the boundaries cleanly. `MPI_PROC_NULL` rather than a special case for every edge.

Part C: Measure and reconcile (11 marks)

C1. Strong-scaling study at $P = 1, 2, 4, 8, 16, 32$, spanning at least two nodes. Speedup, efficiency, Karp-Flatt, with plots.

C2. Weak-scaling study: fix work per rank, grow the problem. Plot runtime against P . Comment on the flatness.

C3. Plot the measured communication fraction against your Part A3 prediction. Do they agree in trend? In magnitude? Name two effects the model omits.

C4. Compare intra-node and inter-node scaling. At what point does crossing a node boundary cost you visibly, and does that match your Lab 6 α and β ?

C5. State the P beyond which adding ranks makes your program **slower**, and explain the mechanism.

What to submit

assign3-<ROLLN01>-<ROLLN02>.pdf plus source. Both names, and a statement of who did what.

MARKS	FOR
7	Part A: a real design with predictions made in advance
12	Part B: correct at awkward rank counts, real overlap, collectives used properly
11	Part C: both studies, and an honest reconciliation of prediction with measurement

NOTE · PAIRS

Both members must be able to explain any part of the submission. Where one cannot, that member is marked accordingly.